JAMES CHRISTOPHER HARPER

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PROFESSIONAL SUMMARY

Forward Deployed Engineer with 7 years of hands-on experience deploying and maintaining infrastructure across remote and austere environments. I've spent the last three years working directly with field teams to solve real-time operational challenges, combining infrastructure expertise with practical problem-solving. My background spans cloud architecture, network deployment, and system hardening, with a particular focus on resilience in disconnected or bandwidth-constrained settings.

WORK EXPERIENCE

Senior Forward Deployed Engineer

Meridian Systems Ltd

meridian-systems.io

London, United Kingdom

March 2022 - Present

Deployed and managed Kubernetes clusters across 12 remote locations, working directly with operations teams to ensure 99.2% uptime despite challenging network conditions and limited technical support on-site.

Designed and implemented automated infrastructure provisioning using Terraform and Ansible, reducing deployment time from 6 weeks to 4 days and eliminating manual configuration errors across distributed teams.

Troubleshot critical system failures in real-time across three continents, coordinating with local technicians via satellite links and offline documentation to restore service without remote access capabilities.

Developed offline-first monitoring and alerting system using Prometheus and custom Python services, enabling field teams to detect and respond to issues without constant cloud connectivity.

Forward Deployed Engineer

Tactical Networks Inc

tacnetworks.co.uk

Berlin, Germany and various deployments

November 2019 - February 2022

Deployed containerised applications to austere environments with minimal infrastructure, including edge computing solutions that operated in areas with no reliable internet access.

Built comprehensive deployment runbooks and operational procedures that enabled junior field technicians with limited Linux experience to maintain complex systems independently.

Implemented network hardening and security protocols across 8 separate deployment sites, conducting security audits and implementing zero-trust architecture principles adapted for bandwidth-constrained networks.

Managed infrastructure cost optimisation by rightsizing cloud resources and implementing intelligent caching layers, reducing monthly cloud expenditure by 34% without compromising performance.

Infrastructure Engineer

CloudBase Technologies

www.cloudbase.com

London, United Kingdom

July 2017 - October 2019

Provisioned and maintained cloud infrastructure on AWS and Azure, supporting 25+ internal teams and managing infrastructure budgets totalling 150k GBP annually.

Implemented CI/CD pipelines using Jenkins and GitLab CI, reducing deployment cycles from 2 hours to 12 minutes and enabling the engineering team to ship updates 20 times daily.

Supported database migration of legacy on-premises systems to PostgreSQL on managed cloud services, planning and executing zero-downtime migrations for three production databases serving 50,000+ users.

Documented infrastructure-as-code practices and led training sessions for 40 junior engineers, establishing standards that became the foundation for company-wide infrastructure management.

EDUCATION

MSc Computer Science

University of Edinburgh

Edinburgh, Scotland

2012 - 2014

BSc Information Technology

University of Manchester

Manchester, England

2008 - 2012

TECHNICAL SKILLS

Cloud Platforms: AWS, Azure, Google Cloud Platform

Container and Orchestration: Docker, Kubernetes, Docker Compose

Infrastructure as Code: Terraform, Ansible, CloudFormation

Monitoring and Logging: Prometheus, Grafana, ELK Stack, Datadog

Programming Languages: Python, Go, Bash, Ruby

Databases: PostgreSQL, MySQL, MongoDB, Redis

Networking: TCP/IP, DNS, BGP, VPN, Wireguard, network troubleshooting

CI/CD: Jenkins, GitLab CI, GitHub Actions

Version Control: Git, GitHub, GitLab

Operating Systems: Linux Ubuntu, CentOS, Alpine, macOS

Security: SSL/TLS, PKI, Linux hardening, vulnerability scanning

GITHUB REPOSITORIES

jharper-deploy/k8s-offline-deployment

Kubernetes deployment toolkit optimised for low-bandwidth and offline environments, featuring automated failover and local caching mechanisms.

jharper-deploy/terraform-austere-infrastructure

Collection of Terraform modules for deploying resilient infrastructure in remote locations with limited connectivity and power constraints.

jharper-deploy/field-ops-monitor

Lightweight monitoring system designed for distributed teams with no cloud access, using statsd and local data storage for systems in disconnected environments.

jharper-deploy/ansible-network-hardening

Ansible playbooks for implementing network security and system hardening across Linux infrastructure in resource-constrained deployments.